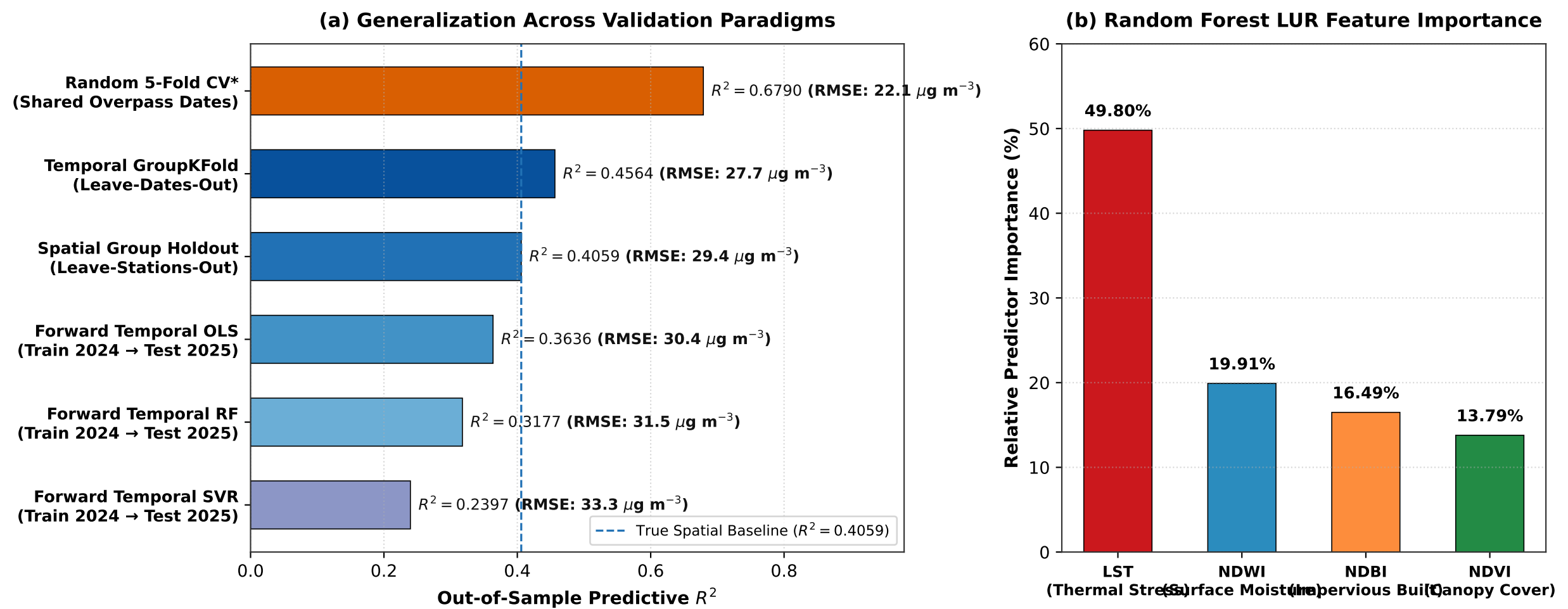
(a) Generalization Across Validation Paradigms
Random 5-Fold CV*
(Shared Overpass Dates)
$R^2 = 0.6790$ (RMSE: 22.1 μg m^{-3})
Temporal GroupKFold
(Leave-Dates-Out)
$R^2 = 0.4564$ (RMSE: 27.7 μg m^{-3})
Spatial Group Holdout
(Leave-Stations-Out)
$R^2 = 0.4059$ (RMSE: 29.4 μg m^{-3})
Forward Temporal OLS
(Train 2024 → Test 2025)
$R^2 = 0.3636$ (RMSE: 30.4 μg m^{-3})
Forward Temporal RF
(Train 2024 → Test 2025)
$R^2 = 0.3177$ (RMSE: 31.5 μg m^{-3})
Forward Temporal SVR
(Train 2024 → Test 2025)
$R^2 = 0.2397$ (RMSE: 33.3 μg m^{-3})
True Spatial Baseline ($R^2 = 0.4059$)
0.0
0.2
0.4
0.6
0.8
Out-of-Sample Predictive R^2
(b) Random Forest LUR Feature Importance
Relative Predictor Importance (%)
0
10
20
30
40
50
60
49.80%
19.91%
16.49%
13.79%
LST
(Thermal Stress)
NDWI
(Surface Moisture)
NDBI
(Impervious Built)
NDVI
(Canopy Cover)